Multi-Objective Archimedes Optimization Algorithm with Fusion-based Deep Learning model for brain tumor diagnosis and classification

DA 243 Presentation

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Outline

1. Introduction
2. Build-up
3. Proposed Framework
4. Conclusion

Introduction

1. The Clinical Problem: Brain Tumors

- **Definition:** A brain tumor (BT) is an abnormal aggregation of cells growing inside brain tissue.
- **Types:**
  - **Benign:** Non-cancerous, often curable via surgery.
  - **Malignant:** Cancerous and potentially fatal.
  - **Primary:** Originates in the brain.
  - **Metastatic:** Spreads from other body parts.
- **Impact Statistics (Global Cancer Analysis 2020):**
  - 308,000 new cases (1.6% of all cancers).
  - 251,000 deaths (2.5% of cancer deaths).
- **Key Need:** Early diagnosis significantly improves survival and treatment planning.

Introduction

2. Current Standard: Medical Imaging & CAD

- **MRI Dominance:**
  - Preferred modality in neurology.
  - Higher soft-tissue contrast than CT.
  - No ionizing radiation exposure.
- **Computer-Aided Diagnosis (CAD):**
  - Acts as a "second opinion" for radiologists.
  - Typical CAD Pipeline:
    1. Preprocessing (Noise removal)
    2. Feature Extraction
    3. Classification (Supervised Learning)

Building Up

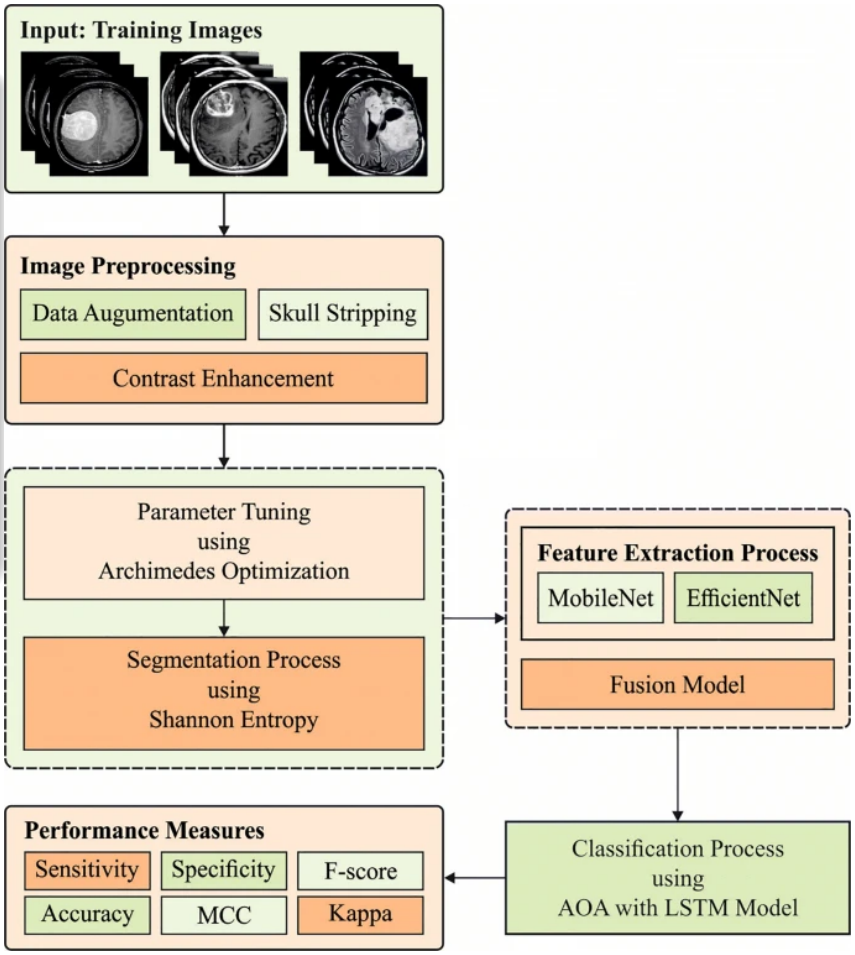
Limitations of Current Diagnostic Methods

- **Inefficient Feature Extraction:**
  - Manual extraction (color, shape, texture) struggles with high-dimensional data.
  - Deep Learning performs better but requires high computational power.
- **Optimization Issues:**
  - Metaheuristic algorithms suffer from premature convergence.
  - Often get trapped in local optima.
- **Hyperparameter Tuning:**
  - Manual tuning (learning rate, batch size, etc.).
  - Tedious and prone to human error.
- **Segmentation Accuracy:**
  - Kapur's and Tsallis's thresholding may struggle with MRI ambiguity.
  - Shannon's entropy can offer improved robustness.

Proposed Framework Overview

Workflow Steps

- Image Preprocessing
- Image Segmentation
- Feature Extraction & Fusion
- Classification Process
- Performance Validation



The Three Pillars of Preprocessing

A. Contrast Enhancement (CLAHE)

- Uses **Contrast Limited Adaptive Histogram Equalization**.
- Enhances contrast in small local regions.
- Controls noise amplification.
- Improves tumor boundary visibility.
- Supports better Shannon entropy–based segmentation.

B. Skull Stripping

- Removes non-brain tissues (skull and background).
- High-intensity pixels (>200) are filtered out.
- Solidity-based filtering eliminates unwanted regions.
- Prevents histogram distortion and improves threshold selection.

The Three Pillars of Preprocessing (Contd.)

C. Data Augmentation

- Prevents overfitting.
- Improves deep learning model generalization.
- Applied to models like **MobileNet** and **EfficientNet**.
- Techniques used:
  - Horizontal and vertical flipping
  - Rotation (90° and 180°)

Image Segmentation – Histogram Representation

A. From Preprocessed Image to Intensity Space

Technical:

Preprocessed MRI image:

$$I(x, y)$$

Gray-level set:

$$G = \{0, 1, 2, \ldots, L-1\}$$

- Each pixel intensity belongs to G.
- The image can be viewed as a distribution over gray levels.

Brain Tumor Interpretation:

- Different tissues (tumor, normal brain, background) have different intensity values.
- These differences appear in the gray-level distribution.

Image Segmentation – Normalized Histogram H

B. Normalized Histogram Representation

Technical:

For gray-level set $G = \{0, 1, \ldots, L-1\}$,

$$h_i = \frac{n_i}{M \times N}, \quad H = \{h_0, h_1, \ldots, h_{L-1}\}, \quad \sum_{i=0}^{L-1} h_i = 1$$

- h_i represents the probability of intensity level i.
- H is the probability distribution of the preprocessed image.
- Segmentation optimization is performed over H.

Brain Tumor Interpretation:

- The spatial image $I(x, y)$ is summarized as distribution H.
- Tumor and healthy tissues create distinct peaks in H.
- Threshold selection corresponds to partitioning H.

AOA Parameters – Decision and Physical Variables

1. Decision Variable and Physical Parameters

Decision Variable:

$$T = (t_1, t_2, \ldots, t_k), \quad T \in [0, L-1]^k$$

- T is the threshold vector to be optimized.

Physical Parameters:

$$den_i = rand, \qquad vol_i = rand$$

$$acc_i = lb + rand(ub - lb)$$

- den_i (density) and vol_i (volume) influence object interaction.
- acc_i (acceleration) determines movement magnitude.

AOA Parameter – Transfer Function

2. Transfer Function

Technical:

$$TF = \exp\left(\frac{t - t_{max}}{t_{max}}\right)$$

- Controls transition from exploration to exploitation.
- If $TF \leq 0.5 \rightarrow$ Exploration phase.
- If $TF > 0.5 \rightarrow$ Exploitation phase.

Interpretation:

- Early iterations $\rightarrow$ global search.
- Later iterations $\rightarrow$ local refinement.

AOA Updating – Exploration Phase

3. Exploration Phase ($TF \leq 0.5$)

Acceleration Update:

$$acc_i^{t+1} = \frac{den_{mr} + vol_{mr} \cdot acc_{mr}}{den_i^{t+1} \cdot vol_i^{t+1}}$$

Position Update:

$$T_i^{t+1} = T_i^t + C \cdot rand \cdot acc_i$$

- Random interaction among objects.
- Large movements across search space.

AOA Updating – Exploitation Phase

4. Exploitation Phase ($TF > 0.5$)

Acceleration Update:

$$acc_i^{t+1} = \frac{den_{best} + vol_{best} \cdot acc_{best}}{den_i^{t+1} \cdot vol_i^{t+1}}$$

Position Refinement:

$$T_i^{t+1} = T_i^t + C \cdot rand \cdot acc_i$$

- Movement guided by best solution found.
- Fine adjustments near optimal threshold vector.

AOA – Optimization Framework

Archimedes Optimization Algorithm

Technical Overview

- Population-based metaheuristic (buoyancy principle).
- Each object represents a candidate threshold vector.

$$T_i = (t_1, t_2, \ldots, t_k)$$

$$T_i = lb + rand(ub - lb)$$

$$TF = \exp\left(\frac{t - t_{max}}{t_{max}}\right)$$

Interpretation: Each T_i defines a possible tumor boundary.

AOA Search Mechanism

Exploration vs Exploitation

Exploration ($TF \leq 0.5$)

$$acc_i^{t+1} = \frac{den_{mr} + vol_{mr} acc_{mr}}{den_i^{t+1} vol_i^{t+1}}$$

Exploitation ($TF > 0.5$)

$$acc_i^{t+1} = \frac{den_{best} + vol_{best} acc_{best}}{den_i^{t+1} vol_i^{t+1}}$$

Position Update

$$T_i^{t+1} = T_i^t + C \cdot rand \cdot acc_i$$

Interpretation: Exploration diversifies solutions, exploitation refines best threshold.

Shannon Entropy as Fitness Function

Entropy-Based Objective

Histogram Representation

$$H = \{h_0, h_1, \ldots, h_{L-1}\}$$

Regional Terms

$$P_j = \sum_{i \in R_j} h_i \qquad H_j = -\sum_{i \in R_j} \frac{h_i}{P_j} \log\left(\frac{h_i}{P_j}\right)$$

Total Objective

$$H(T) = \sum_{j=1}^{k+1} H_j$$

Optimization Problem

$$T^* = \arg \max_{T \in [0, L-1]^k} H(T)$$

Why Use AOA for Image Segmentation?

C1. Justification for Using AOA

Technical:

- Objective function $H(T)$ is nonlinear and non-convex.
- Search space: $T \in [0, L-1]^k$ with constraint $t_1 < t_2 < \cdots < t_k$.
- Gradient-based methods cannot be applied.
- Brute-force search is computationally expensive.
- AOA provides global search with exploration–exploitation balance.

Comparison:

- Gradient Descent $\rightarrow$ may get stuck in local maxima.
- Exhaustive search $\rightarrow$ exponential complexity.
- AOA $\rightarrow$ efficient global optimization without derivatives.

Brain Tumor Interpretation:

- MRI histograms are multi-modal and irregular.

Fusion-Based Feature Extraction

EfficientNet-B0

- 18 convolution layers
- Kernel size: 3×3 / 5×5
- Input: RGB, 224×224
- Transfer learning used

MobileNet

- Depthwise (3×3) + Pointwise (1×1) convolution
- Reduced parameters
- BatchNorm + ReLU activation

Feature Fusion and Selection

Deep Feature Extraction

EfficientNet feature vector:

$$f_{EFF1n} = \{EFF_{11}, EFF_{12}, \ldots, EFF_{1n}\}$$

MobileNet feature vector:

$$f_{MN1p} = \{MN_{11}, MN_{12}, \ldots, MN_{1p}\}$$

where n and p denote feature dimensions extracted from EfficientNet-B0 and MobileNet, respectively.

Feature Fusion

The extracted features are fused into a single vector:

$$\text{Fused Feature Vector } (1 \times q) = \sum_{i=1}^{2} f_{EFF1n}\ f_{MN1p}$$

Final fused feature dimension:

$$f = (1 \times 1186)$$

Feature Fusion and Selection

Entropy-Based Feature Selection

Entropy-based scoring:

$$B_{He} = NH_e^b \sum_{i=1}^{n} p(f_i)$$

Top 1186 features are selected based on Entropy:

$$F_{select} = \max\{f_i\} \ (1186)$$

where H_e denotes entropy and $p(f_i)$ represents feature probability.

Image Classification: LSTM

Why LSTM?

- Handles sequential dependencies
- Avoids vanishing gradient issue
- Suitable for deep feature sequences

Gate Equations

$$f_t = \sigma(W_f[h_{t-1}, x_t] + B_f)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + B_i)$$

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

$$h_t = o_t \tanh(C_t)$$

$$Output = \text{softmax}(h_t)$$

AOA-Based Hyperparameter Optimization

Hyperparameter Vector

For LSTM classifier, define:

$$x_i = \{\eta,\ B,\ E\}$$

- η : Learning rate
- B : Batch size
- E : Number of epochs

Each AOA object represents one candidate hyperparameter set.

Fitness Function

Objective: Minimize classification error

$$\text{Fitness}(x_i) = \frac{\text{Misclassified Samples}}{\text{Total Samples}} \times 100$$

Lower fitness $\Rightarrow$ Better hyperparameters

AOA Search for Optimal Hyperparameters

Initialization

$$x_i = lb + rand(ub - lb)$$

$$den_i = rand, \quad vol_i = rand, \quad acc_i = lb + rand(ub - lb)$$

Update Strategy

Transfer function:

$$TF = \exp\left(\frac{t - t_{max}}{t_{max}}\right)$$

- $TF \leq 0.5 \rightarrow$ Exploration
- $TF > 0.5 \rightarrow$ Exploitation

Best solution:

$$x^* = \arg\min \text{Fitness}(x_i)$$

Why Use AOA for Hyperparameter Optimization?

Optimization Perspective

- Hyperparameter search space is nonlinear
- Gradient-based tuning is not applicable
- Manual tuning is tedious and unstable
- Grid search is computationally expensive

Advantages of AOA

- Global search capability
- Balanced exploration–exploitation
- Fast convergence
- Avoids local minima

Performance Validation

Experimental Setup & Metrics

Datasets: Kaggle (253 MRI images: 98 Normal, 155 Tumor), Figshare (3064 MRI slices, 3 tumor types).
Data Split: 70% Training, 30% Testing.
Evaluation Metrics: Sensitivity, Specificity, Accuracy, F-Score, MCC, Kappa, CT.

$$\text{Fitness} = \frac{\text{Misclassified}}{\text{Total}} \times 100$$

Results

Kaggle: Accuracy 98.95%, Sensitivity 97.96%, Specificity 99.57%.
Figshare: Accuracy 99.80%, Kappa 99.56%.
Computational Time: 4.32s (Kaggle), 5.20s (Figshare), faster than SVM, ResNet50, Inception V3.

Conclusion

Research Summary

- Proposed a **Multi-Objective Archimedes Optimization based Fusion Deep Learning (MOAOA-FDL)** framework for brain tumor diagnosis.
- Integrated: preprocessing (CLAHE + skull stripping), entropy-based multi-level segmentation, feature fusion (MobileNet + EfficientNet), and LSTM classification.
- Modeled segmentation as a nonlinear entropy maximization problem:

$$T^* = \arg\max H(T)$$

- Utilized AOA for both threshold optimization and hyperparameter tuning.
- Experimental results demonstrate superior accuracy and robustness on benchmark MRI datasets.

References

Optimization and Segmentation

[1] L. Abualigah et al., "The Arithmetic Optimization Algorithm," *Comput. Methods Appl. Mech. Eng.*, 376, 113609, 2021.

[2] V. Rajinikanth et al., "Entropy Based Segmentation of Tumor from Brain MR Images," *Pattern Recognition Letters*, 94, 87–95, 2017.

[3] B. Devanathan and K. Venkatachalapathy, "Optimal Kapur's Thresholding with DNN," *IIOABJ*, 11(4), 2020.

Deep Learning Architectures

[4] M. Tan and Q. Le, "EfficientNet: Rethinking Model Scaling for CNN," *ICML*, 2019.

[5] Y. Chiu et al., "MobileNet-SSDv2: Improved Object Detection," *IEEE ICSSE*, 2020.

[6] P.N. Srinivasu et al., "Skin Disease Classification using MobileNet V2 and LSTM," *Sensors*, 21(8), 2021.

References

Brain Tumor Classification Studies

[7] T. Sadad et al., "Brain Tumor Detection and Multi-classification using DL," *Microscopy Research and Technique*, 2021.

[8] M.I. Sharif et al., "Decision Support System for Brain Tumor Classification," *Complex & Intelligent Systems*, 2021.

[9] M. Woźniak et al., "Deep Neural Network Correlation Learning for CT Brain Tumor Detection," *Neural Computing and Applications*, 2021.

References (Datasets)

Datasets Used

- Brain MRI Images for Brain Tumor Detection . Kaggle.
  https://www.kaggle.com/datasets/navoneel/brain-mri-images-for-brain-tumor-detection

- Brain Tumor Dataset (3,064 T1-weighted contrast-enhanced MRI scans) . Figshare.
  https://figshare.com/articles/dataset/brain_tumor_dataset/1512427